

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA02

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analysing images.. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 3 Total Marks:30 Annexure A

Scenario-Based

Code of Conduct:

I SHAIK ALI BABU (192524038) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

			Scenario based assessment			
Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	

Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Annexure A

Scenario-Based

1. Crack Detection on Concrete Surfaces Using Drone Images

Scenario: A civil inspection team is developing a computer vision system to detect thin cracks on concrete bridges and building surfaces. Images are captured using a drone under outdoor conditions. Due to changing sunlight, shadows, and camera vibration, the raw images contain uneven illumination, low contrast, and Gaussian noise.

Questions:

A. Propose a suitable image preprocessing pipeline to prepare the drone images for crack detection. Your answer should include methods for illumination correction, noise reduction, contrast enhancement, and image normalization. Justify the need for each step.

B. Select an appropriate edge detection technique for identifying thin surface cracks. Compare it with at least one other edge detection method and explain why your selected technique is more suitable for detecting fine cracks.

C. After edge detection, some crack segments appear broken or disconnected. Suggest a suitable post-processing technique to connect the crack regions and improve the final detection output.

A. Image Preprocessing Pipeline for Crack Detection

To detect thin cracks accurately, the drone images should be processed before edge detection. A suitable preprocessing pipeline is given below:

Step 1: Illumination Correction

- Method: Adaptive Histogram Equalization (CLAHE) or Homomorphic Filtering.
- Reason: Drone images captured outdoors often contain shadows and uneven sunlight. Illumination correction makes the brightness more uniform across the image so that cracks remain visible in both bright and dark areas.

Step 2: Noise Reduction

- Method: Gaussian Filter or Bilateral Filter.
- Reason: Camera vibration and environmental conditions introduce Gaussian noise. Noise reduction removes unwanted pixels while preserving important crack structures.

Step 3: Contrast Enhancement

- Method: Contrast Limited Adaptive Histogram Equalization (CLAHE).
- Reason: Thin cracks usually have very low contrast compared to the concrete surface. Contrast enhancement increases the visibility of cracks without over-amplifying noise.

Step 4: Image Normalization

- Method: Min-Max Normalization or Z-score Normalization.
- Reason: Normalization scales pixel intensity values to a common range. This improves consistency between different images captured under different lighting conditions and helps the edge detector perform more accurately.

Preprocessing Pipeline

Input Image → Illumination Correction → Noise Reduction → Contrast Enhancement → Image Normalization → Edge Detection

B. Suitable Edge Detection Technique

Selected Method: Canny Edge Detection

The Canny edge detector is the best choice because it can accurately detect very thin cracks while reducing false edges caused by noise.

Advantages of Canny Edge Detection

- Detects weak and thin edges effectively.
- Includes noise smoothing before edge detection.
- Uses double thresholding to reduce false detections.
- Produces continuous and well-defined edges.

Why Canny is Better

Concrete cracks are usually narrow and faint. The Canny algorithm preserves these fine details while suppressing noise, making it more reliable than the Sobel operator.

C. Post-Processing Technique

Morphological Closing

After edge detection, crack segments may appear disconnected because of shadows or image noise.

Method

- **Apply Morphological Closing (Dilation followed by Erosion).**

Benefits

- Connects nearby crack segments.
- Fills small gaps.
- Produces continuous crack boundaries.
- Improves the final crack detection result.

2. Human Motion Tracking in a Security Camera Video

Scenario: A security camera is installed in a hallway to monitor people moving in different directions. The system must estimate the motion of each person across video frames for tracking and activity analysis. The hallway contains plain walls, smooth floors, and occasional shadows.

Questions:

- A.** Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.
- B.** Discuss how the aperture effect can create errors in motion estimation, especially on uniform regions such as plain walls, floors, and clothing surfaces.
- C.** If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy in such situations.

A. Lucas-Kanade Optical Flow

The **Lucas-Kanade Optical Flow** method estimates the movement of objects between two consecutive video frames.

Working

1. Detect feature points such as corners.
2. Assume the motion within a small image window is nearly constant.
3. Compare the position of each feature point in consecutive frames.
4. Calculate the displacement vector that represents the object's movement.

Application

- Tracks people while they move through the hallway.
- Estimates movement direction and speed.
- Supports activity analysis and continuous tracking.

B. Aperture Effect

The **aperture effect** occurs when only a small portion of an object is visible.

Problems in Uniform Regions

- Plain walls have very few texture features.
- Smooth floors provide little information for motion estimation.
- Clothing with uniform colors also lacks identifiable features.

As a result:

- Motion direction becomes uncertain.
- Optical flow vectors may be incorrect or incomplete.
- Tracking accuracy decreases.

C. Sudden Change in Speed or Direction

When a person suddenly changes direction or speed:

- Optical flow vectors change rapidly.
- Vector lengths increase or decrease depending on speed.
- Vector directions shift according to the new movement.

These sudden changes may temporarily reduce tracking accuracy.

Improvement Method

Use a **Kalman Filter**.

Advantages

- Predicts the person's next position.
- Handles sudden motion changes smoothly.
- Reduces tracking errors caused by temporary motion variations.
- Improves overall tracking stability.

3. Vehicle Tracking at a Traffic Junction Using Video Frames

Scenario: A smart traffic monitoring system uses CCTV footage to track vehicles at a busy road junction. Vehicles move at different speeds, some stop suddenly at signals, and others turn left or right. The video also contains shadows, reflections, and partial occlusion when vehicles overlap.

Questions:

A. Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

B. Identify two challenges that may affect accurate motion estimation in this scenario, such as occlusion, shadows, sudden braking, or camera vibration. Explain how each challenge affects the optical flow result.

C. Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly.

A. Optical Flow for Vehicle Tracking

Optical flow estimates the movement of vehicles by analyzing changes between consecutive video frames.

Working

- Detect moving vehicle features.
- Compute displacement between frames.
- The **direction** of the optical flow vector indicates vehicle movement.
- The **length** of the vector represents vehicle speed.

Applications

- Estimate traffic speed.
- Detect vehicle movement direction.
- Monitor traffic flow automatically.

B. Challenges Affecting Motion Estimation

1. Occlusion

When one vehicle passes in front of another:

- Parts of the hidden vehicle disappear.
- Optical flow vectors become incomplete.
- The tracking system may lose the vehicle temporarily or confuse two vehicles.

2. Shadows

Vehicle shadows also move with the vehicles.

Effect

- Shadows may be detected as moving objects.
- Incorrect optical flow vectors are generated.
- Tracking accuracy decreases because shadows are mistaken for actual vehicles.

C. Improvement Method

Kalman Filter with Object Detection

A combination of **Kalman Filter** and object detection improves tracking reliability.

Benefits

- Predicts vehicle positions during temporary occlusion.
- Continues tracking even when vehicles overlap.
- Handles sudden turns and braking smoothly.
- Reduces tracking errors caused by missing detections.